

TITLE PAGE

- Problem Statement ID - **DJS_26_SW_07**
- Problem Statement Title - **Automated System for Career Advancements of the Faculties of Higher Education**
- PS Category - **Software**
- Team ID - **DJSW_01**
- Team Name - **GitGoneWild**



Detailed Explanation of the Proposed Solution:

- **Unified Academic Career Record**- Maintain **one verified academic activity record** with linked evidence for all academic work.
- **Continuous Activity Capture** - Automatically collect signals from **emails, calendars, drives, publications, CVs, events, and manual entries throughout the year.**
- **Intelligent Structuring & Confidence** - **Normalize, classify, deduplicate, and score detected activities** using deterministic rules with AI only where needed.
- **Faculty-Controlled Verification** - Treat **every detected activity as a proposal** until the faculty member reviews its evidence and confirms it.
- **One Record → Institution-Wide Outputs** - Reuse **verified records to generate CVs, appraisals, promotion dossiers, forms, and departmental reports.**

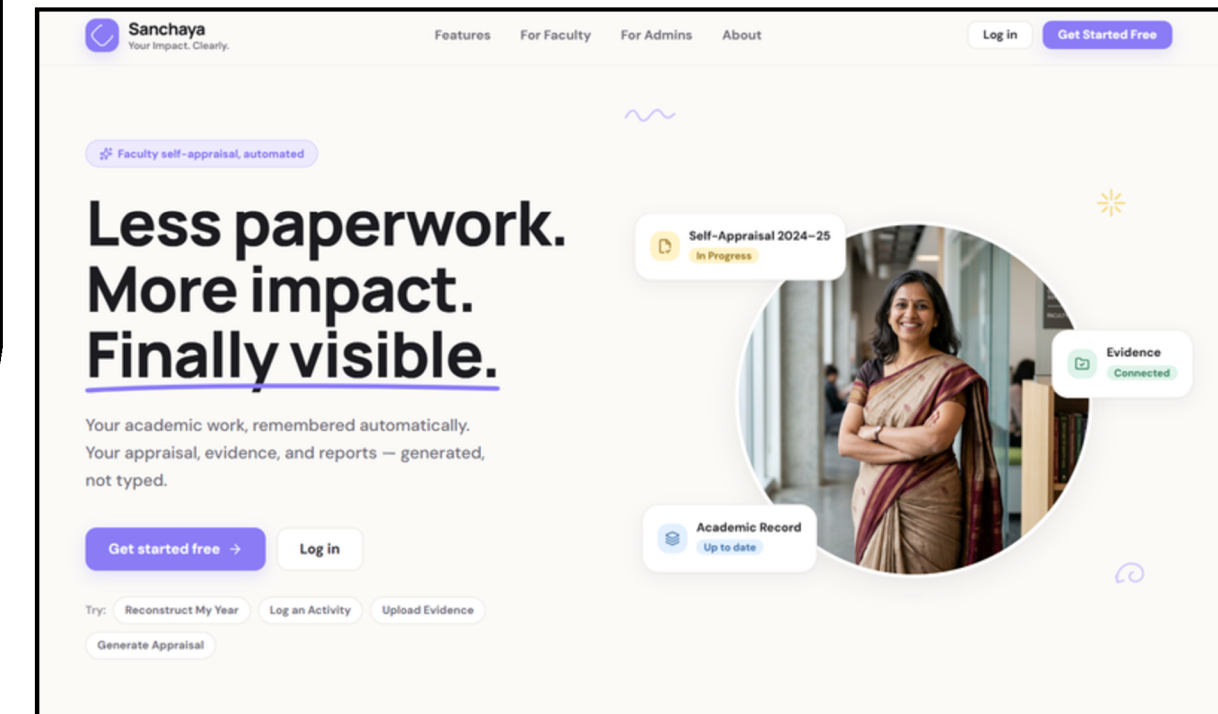
Innovation and Uniqueness:

- **Continuous academic memory** - build the faculty record **throughout the year**, not during appraisal season.
- **Evidence-backed activities** with **linked sources** for every detected contribution.
- **One record, multiple outputs** - generate CVs, appraisals, dossiers and reports instantly.
- **Never automate silently** - human approval before anything becomes official.
- **Built beyond annual appraisal** - the same foundation supports career progression and academic mobility.

How it addresses the problems:

- Reduces **appraisal workload** by capturing activities before review season.
- Eliminates **repetitive data entry through automatic discovery** and deduplication.
- Centralizes **scattered evidence** into one searchable academic repository.
- **Captures** overlooked contributions such as mentoring, reviewing, committees and service.
- **Simplifies administration** through automated forms, promotion tracking and reporting.

Step Inside Sanchay.
Experience the academic career platform live.



Deployed Site

Demo Video ↗

1) Academic Data Ingestion

- o ORCID, Gmail, Calendar, Drive, CVs, certificates.
- o **Normalize** → **classify** → **deduplicate** incoming records.

2) Activity Discovery & Correlation

- o Extract **event/date/role/organization/evidence** automatically.
- o **Time-stamped audit logs** and versioning.

3) Faculty Verification

- o Every detected activity remains a **proposal**.
- o **Confirm** → **Edit** → **Ignore** before official entry.

4) Canonical Academic Record :

- Store verified activities + linked evidence **once**.
- Maintain one **continuous academic career** record.

5) Automated Document Generation

- o Generate **appraisals, forms, CVs, dossiers, reports**.
- o Reuse verified data across different templates.

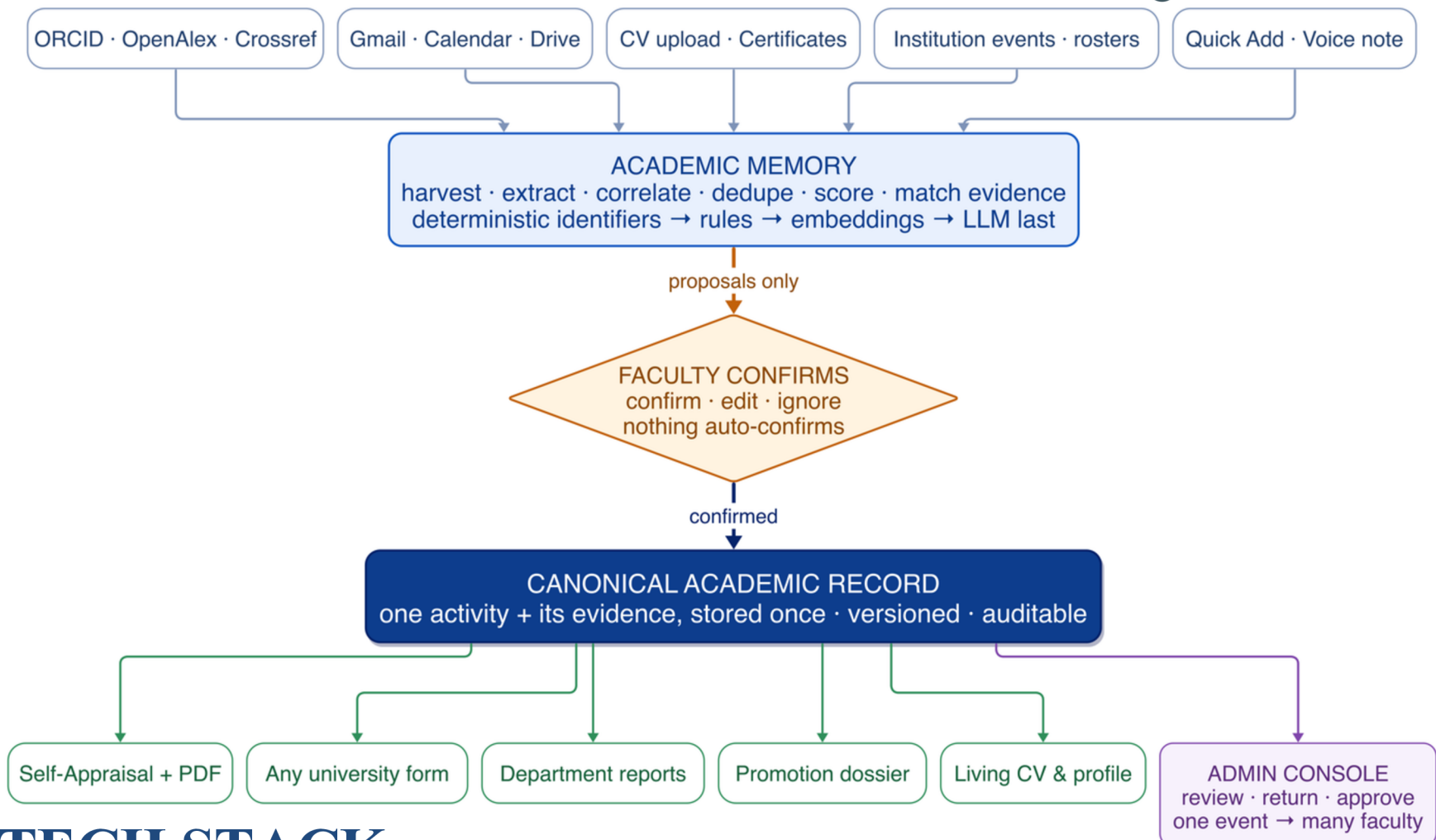
6) Institutional Coordination

- o Real-time **faculty** ↔ **admin** review workflows.
- o One institutional event can update many faculty.

7) Security & Scalability

- o RBAC, RLS, encrypted integrations, signed files.
- o **Async** processing + **indexed DB** for large-scale deployment.

APPLICATION FLOW



TECH STACK :

Frontend: Next.js 15, TypeScript, Tailwind CSS

Backend: FastAPI, Python 3.12, Pydantic, SQLAlchemy

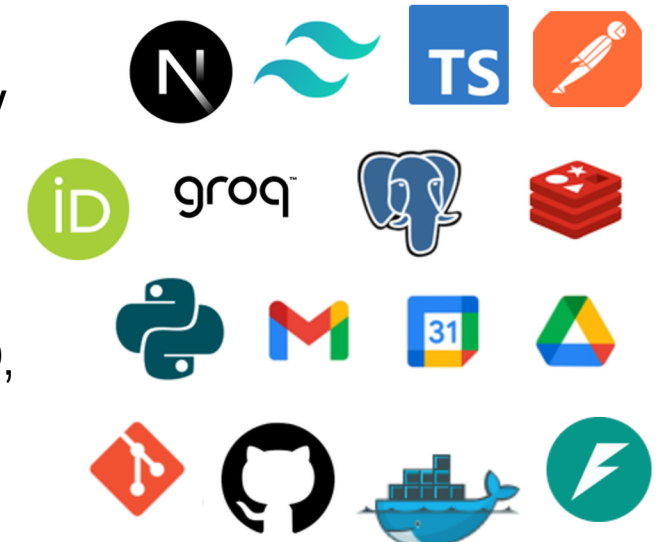
Database: PostgreSQL (Supabase), pgvector, Redis

AI & Automation: Structured LLM Pipelines, Docling, PyMuPDF, ARQ, GPT-OSS 120B, Qwen Vision

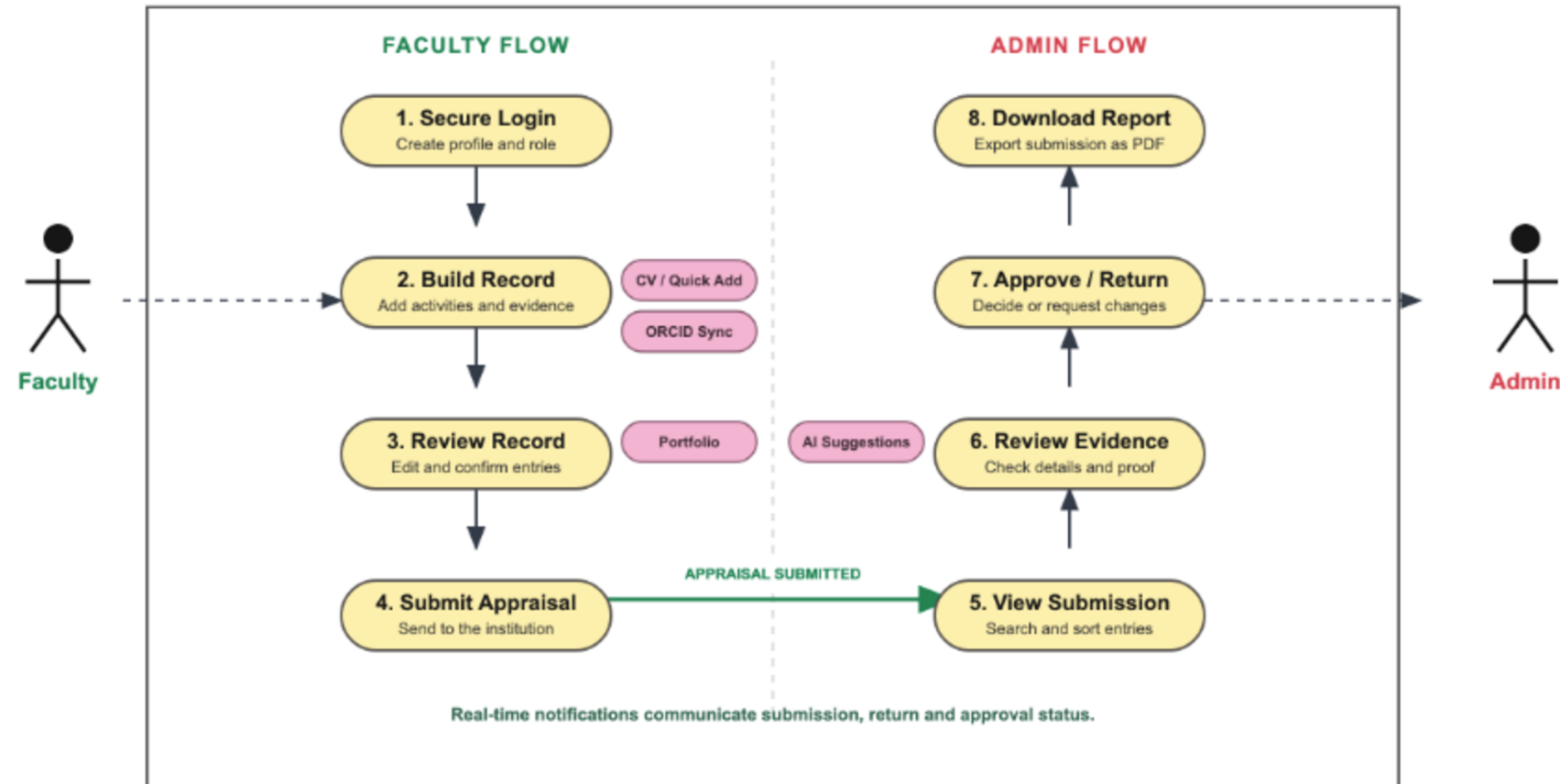
Integrations: Gmail, Calendar, Drive, Postman, ORCID, OpenAlex, Crossref

Infrastructure: Docker, Vercel, Railway, Upstash

Version Control : Git, Github



PROVISIONED USER FLOW



Potential Challenges and Risks

- 1) Multilingual Adaption:** Current systems work with **English** data; global expansion requires region-specific language, rules and support.
- 2) Bias in automated recommendations:** Promotion **criteria**, scoring **metrics**, appraisal **readiness** may vary with institution, region, beliefs.
- 3) Digital Literacy Barriers: Evolving** Smaller colleges with poor internet access, limited IT infra, heavy paperwork may hesitate to onboard.
- 4) Cybersecurity threats & Infra. :** **Phishing** attacks, **Data breaches**, **takeovers** can cause universities to fear losing/ leaking data. **Storage**, **APIs**, can raise costs.

Feasibility Analysis

- 1) Economic Feasibility:** **Low cost** cloud services make a department level **pilot** financially practical.
- 2) Technical Feasibility:** **Live platform** at **pre-production** test proves the solution can be built and operated reliably.
- 3) Operational Feasibility:** **Quick Add**, **CV import**, **publication sync**, **evidence storage**, **automated forms** and **real-time admin review** reduce repetitive data entry, paperwork, follow-ups and appraisal **processing time**.

Strategies for Overcoming Challenges

- 1) Introduce verifiable digital credentials:** Integrate **DigiLocker** in India, use **Web3/Blockchain** based verification & **partner** with **foreign universities** to ensure legal compliance, **accreditation** and **data-privacy** requirements.
- 2) Vocal for Local Support:** Use **AI** to **transcribe**, **translate** and **map** entries initially. Roll out language support in **phases** with **beta testing**. Enable **voice command**, **dictation** for instructions support.
- 3) Build privacy & compliance by design:** Use explicit **consent**, **read-only** permissions, encrypted **tokens**, **RBAC**, audit **logs** and regional data - retention policies for **DPDP**, **GDPR** and other regulations.

Potential Impact of the Solution

- 1) Lifelong Faculty Record:** Builds a **continuous academic identity** that grows with every achievement, role, and career milestone.
- 2) Continuous Academic Intelligence:** Converts fragmented activities, documents, publications, and evidence into a **structured career record**.
- 3) Evidence-Driven Institutions:** Enables **reliable academic decisions** through structured, verified, and continuously updated faculty information.
- 4) Portable Academic Achievements:** Keeps accomplishments **accessible and reusable** across institutions, roles, appraisals, and career transitions.
- 5) Connected Academic Infrastructure:** Creates a **scalable foundation** linking faculty careers, institutional workflows, research, and professional growth.

Benefits of the Solution

- 1) Capture Once, Reuse Everywhere:** One verified record powers appraisals, reports, forms, CVs, grants, and dossiers.
- 2) Intelligent Workflow Automation:** Automates evidence organization, appraisal preparation, reporting, and repetitive academic administration.
- 3) Opportunity-Led Career Growth:** Surfaces relevant grants, publications, collaborations, research opportunities, and career pathways.
- 4) Connected Academic Ecosystem:** Brings faculty, researchers, mentors, collaborators, and administrators onto one professional platform.
- 5) Real-Time Academic Visibility:** Provides an **always-current view** of progress, achievements, evidence, opportunities, and institutional activity.

RESEARCH AND REFERENCES

Category	Platform / Reference	Relevance to our Solution
Government Infrastructure	VIDWAN Portal - INFLIBNET	Demonstrates a national faculty-profile and expert-discovery network . It supports Sanchaya's vision of reusable faculty profiles, expertise visibility and institutional networking.
Government Infrastructure	API Setu / DigiLocker - MeitY	Reference for consent-based retrieval and verification of government-issued documents. Future integration could verify certificates and reduce manual evidence uploads.
Government Infrastructure	AISHE Portal – Ministry of Education	Provides the national model for structured institutional and faculty-data reporting . Sanchaya can reduce repeated data collection for institutional reporting.
Government Infrastructure	Samarth eGov Suite	A government-backed university ERP covering employee, academic, governance and research processes. It provides a reference for future institutional interoperability.
Indian Regulatory Framework	UGC Performance-Based Appraisal System and CAS	Defines the Indian basis for faculty appraisal, documentation, API/PBAS assessment and Career Advancement Scheme evaluation . Sanchaya digitises this evidence-driven process.
Indian Research Project	Digital Performance-Based Appraisal System – JSSAHER	Validates that digital PBAS can reduce time, improve satisfaction and support automatic performance evaluation. It provides direct research evidence for Sanchaya's feasibility.
International System	CRISTin - Norway's Research Information System	Historical reference for nationally consolidating publications, projects and institutional reporting . It demonstrates how one record can serve researchers, universities and government.
International System	Norwegian Research Information Repository - NVA	CRISTin's current successor . It combines national research reporting, profiles, publication metadata and open-access repositories while reducing duplicate data entry.
Research Identity	ORCID	Provides globally unique researcher identifiers and permission-based record updates . Sanchaya uses it to discover publications and reduce repeated faculty data entry.
Research Intelligence	OpenAlex	Provides a global research index covering works, authors, institutions, topics and citations . It supports publication enrichment and research-profile development.
Publication Verification	Crossref REST API	Supplies DOI-based bibliographic metadata directly from publishers and trusted sources. It helps Sanchaya validate, enrich and deduplicate publication records.
Security and Data Platform	Supabase PostgreSQL, Auth, Storage and Realtime	Provides the technical reference for secure authentication , row-level faculty-data isolation , private evidence storage and real-time review notifications .
Private Indian Systems	Cyborg ERP PBAS and Similar Players	Used as a competitor benchmark for automated scoring, UGC-aligned forms , digital documentation and institutional dashboards . Sanchaya differentiates through AI-assisted continuous record building.